SHEET 3

SIC/XE Machine

Review on SIC/XE

Registers:

- The following additional registers are provided by SIC / XE:

Mnemonic	Number	Special use
B	3	Base register; used for addressing
S	4	General working register-no special use
T	5	General working register-no special use
F	6	Floating-point accumulator (48 bits)

Instruction formats:

Format 1 (1 byte): no memory ref.

8
op

Format 2 (2 bytes): no memory ref. / register-to-register

operations 8	4	4
op	r1	R2

Format 3 (3 bytes):

6	1	1	1	1	1	1	12
op	n	i	x	b	p	e	disp

Format 4 (4 bytes):

6	1	1	1	1	1	1	20
op	n	i	x	b	p	e	address

- Bit *e* is used to distinguish between format 3 and 4 (*e=0* means Format 3, *e=1* means Format 4). [**"+"** is added to the instruction]

Addressing modes:

HOW TO CALCULATE?? **"Bits b,p,x"**

Mode	Indication	Target address calculation
Base relative	b=1,p=0	TA=(B)+disp (0<=disp<=4095)
Program-counter relative	b=0,p=1	TA=(PC)+disp (-2048<=disp<=2047))
Direct addressing	b=0,p=0	TA=disp

- Any of these addressing modes can also becombined with indexed addressing if bit x is set to 1 the term (x) is added in the target address

HOW TO USE??

"Bits I,n"

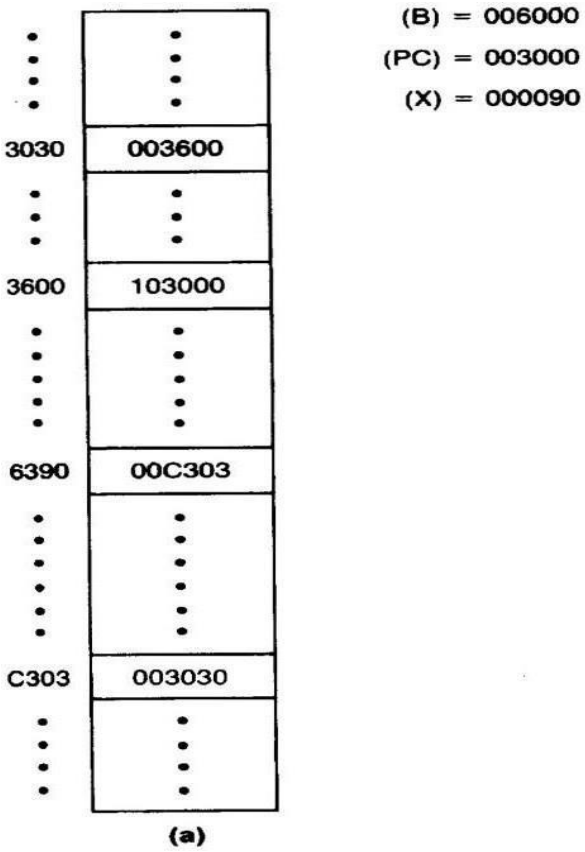
Immediate "#"	i=1,n=0	Value= TA
Indirect Addressing "@"	i=0,n=1	Value=((TA))
Simple	i=1,n=1	Value= (TA)

- Indexing cannot be used with immediate or indirect addressing mode.

- If bits n and i are both 0, then bits b, p, and e are considered to be part of the address field of the instruction (rather than flags indicating addressing mode).This makes Instruction Format 3 identical to the format used on the standard version of SIC, providing the desired compatibility

1)

Memory



Given the following memory structure calculate the target address and find the value loaded into register A for the following machine instructions:

1. 032600
2. 03C300
3. 022030
4. 010030
5. 0310C303

2)

Memory

ABC	6060
2500	ABC
5AEC	30
6060	1090
89AB	C303

(PC)= (2000)

(B)= (5000)

(X)= (0030)

Given the following memory structure calculate the target address and find the value loaded into register A for the following machine instructions:

1. 03A4D0
2. 031089AB
3. 010155
4. 022500
5. 02100ABC
6. 03CABC